

Joseph Agarwal

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SKILLS

Languages: Python, C#, Java, C++, Go, JavaScript, SQL

Tools: Git, .NET, React, Angular, Node, Kafka, Docker, Kubernetes, Jenkins

EXPERIENCE

Tesla

Jan 2025 – Aug 2025

Software Engineer Intern

Fremont, CA

- Engineered a ML model and LLM pipeline to predict and automatically order car parts based on customer repair needs, using **FastAPI**, **AWS S3**, and **LightGBM** to reduce triage costs by **\$2,100,000** monthly
- Led development of an automated email/phone notification system with **ASP.NET**, **C#** and **Cron** to proactively alert managers of upcoming deadlines, improving customer satisfaction scores by **20%** across 192 sites
- Added multilingual speech recognition to note-taking application used by **10,000+ users**, utilizing **TypeScript** and **Angular** to configure auto-fill user mapping, dynamic reformatting, and **Redis** tracking to store user session data
- Spearheaded redesign of legacy microservices to integrate Optimus robotics; achieved a **67%** decrease in code redundancy with **dotnet containerization** and factory dependency injection, automating CI/CD pipelines with **Jenkins** and **Docker**

Powerex Corporation

May 2023 – Aug 2023

Software Engineering Intern

Vancouver, BC

- Constructed **C#** reconciliation app to appraise time-sheet data and generate irregularity reports, reducing weekly manual resourcing time from **2 hours to 2 seconds**, monitored via **Splunk** and **Kubernetes**
- Modernized trading application by leading frontend transition from **AngularJS** to **ReactJS**, equipping **OAuth** authentication and **Redux** session management to increase platform speed by **30%**
- Crafted end-to-end **SOAP** downloader and **API** with certificate credentials and audit logging, saving data to **Snowflake DB** in order to generate **20** customizable reports used by 15+ analysts daily

Cascade Aerospace

Sept 2022 – Apr 2023

Full Stack Developer Intern

Abbotsford, BC

- Engineered a full-stack aeronautical scheduler with **C#** and **JavaScript** by architecting **.NET Core** server and client **REST APIs**, automating allocation workflows and saving **20+ hours** weekly for the department
- Reverse-engineered web-scraping transfer process to standardize data intake, extracting and saving user metadata to **ERP** and **Microsoft SQL Server** databases with **Entity Framework**, tracking and sending 10M data-points weekly

PROJECTS

Stock Backtest Engine | *Python, Node, Flask, AWS, React, JavaScript*

Jan 2024 - Present

- Built intuitive React dashboard with modular hooks and Bootstrap to display Black-Litterman stock projections
- Implemented **multi-regression** analysis using **pandas**, **numpy** and **MySQL** to compute 5 financial performance metrics
- Deployed **Flask** API with **AWS RDS** backend for seamless cloud integration, processing 500+ calls per second

Distributed Warehouse Store | *Java, Citus, SQL, Python*

Aug 2024 - Jan 2025

- Architected and deployed a distributed **PostgreSQL** solution with **Citus**, using advanced sharding across a 5-node cluster to manage high-volume warehouse and shipping data, utilizing **SLURM scripting** for job scheduling
- Reduced SQL query latency by 30% by implementing collocated and repartition join strategies, applying **distributed concurrency control** (2PL) to maintain serializability

Internet Protocol Clone | *C, GDB, netcat, Bash*

Jan 2024 - April 2024

- Utilized **C** to successfully replicate **TCP** and **UDP** functions for a three-way handshake connection and termination.
- Ensured RFC guideline accuracy by implementing header checksums, sender/receiver windows and network timeouts, interpreting SYN, ACK and FIN packet flags, resulting in 40% less retransmissions.

EDUCATION

University of British Columbia

GPA: 3.8

Combined Degree in Computer Science and Business - Dean's Honor List

2020 - 2025

- Awards:** Commerce Undergraduate Society Award in Business, Alexander Rutherford Scholarship
- Leadership:** UBC Trading Group Quant Lead, IMUNA Undersecretary General, VP - UBC Social Enterprise
- Coursework:** Distributed Databases, Quantum Computing, Operating Systems, Data Structures & Algorithms, Networks